C Piscine

Shell 01

Summary: This document is the subject for the module Shell 01 of the C Piscine @ 42.

Version: 7.4

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Chapter I

Instructions

- Only this page will serve as reference; do not trust rumors.
- Watch out! This document could potentially change up before submission.
- These exercises are carefully laid out by order of difficulty - from easiest to hardest. We `will not` take into account a successfully completed harder exercise if an easier one is not perfectly functional.
- Make sure you have the appropriate permissions on your files and directories.
- You have to follow the submission procedures for every exercise.
- Your exercises will be checked and graded by your fellow classmates.
- On top of that, your exercises will be checked and graded by a program called Moulinette.
- Moulinette is very meticulous and strict in its evaluation of your work. It is entirely automated and there is no way to negotiate with it. So if you want to avoid bad surprises, be as thorough as possible.
- Exercises in Shell must be executable with /bin/sh.
- You cannot leave any additional file in your directory than those specified in the subject.
- Got a question? Ask your peer on the right. Otherwise, try your peer on the left.
- Your reference guide is called `Google / man / the Internet / ....`
- Check out the "C Piscine" part of the forum on the intranet.
- Examine the examples thoroughly. They could very well call for details that are not explicitly mentioned in the subject...

Chapter II

Foreword

Here's what Wikipedia has to say about otters :

The European otter (Lutra lutra), also known as the Eurasian otter, Eurasian river otter, common otter and Old World otter, is a European and Asian member of the Lutrinae or otter subfamily, and is typical of freshwater otters.

The European otter is a typical species of the otter subfamily. Brown above and cream below, these long, slender creatures are well-equipped for their aquatic habits. Its bones show osteosclerosis, increasing their density to reduce buoyancy.
This otter differs from the North American river otter by its shorter neck, broader visage, the greater space between the ears and its longer tail.

However, the European otter is the only otter in its range, so it cannot be confused for any other animal. Normally, this species is 57 to 95 cm (23-37 in) long, not counting a tail of 35-45 cm (14-18 in).
The female is shorter than the male.

The otter's average body weight is 7 to 12 kg (15.4-26.4 lbs), although occasionally a large old male may reach up to 17 kg (37 lbs).
The record-sized specimen, reported by a reliable source but not verified, weighed over 24 kg (53 lbs).

The European otter is the most widely distributed otter species, its range including parts of Asia and Africa, as well as being spread across Europe. Though currently believed to be extinct in Liechtenstein, and Switzerland, they are now very common in Latvia, along the coast of Norway and across Great Britain, especially Shetland, where 12% of the UK breeding population exist. Ireland has the highest density of Eurasian otters in Europe.
In Italy, they can be found in southern parts of the peninsula.
The South Korean population is endangered.

Otters are cute.

Chapter III

Exercise 00: Exam

	Exercise : 00
Exam	

- During the week, you will be able to sign up for Friday's exam in the agenda, don't forget.
- You also have to register for the Exam00 project.
- Make sure you've registered for the exam (the event AND the project !).
- Make sure you've made sure you've registered for the exam (the event AND the project ! Yep, both !).

Chapter IV

Exercise 01: print_groups

	Exercise 01
print_groups.sh	
Turn-in directory : *ex*01/	
Files to turn in : `print_groups.sh`	
Allowed functions : `None`	

- Write a command line that will display the list of groups for which the login, contained in the environment variable FT_USER , is a member. Separated by commas without spaces.
- Examples :
  - for FT_USER=nours, the result is "god,root,admin,master,nours,bocal" (without quotation marks)

    ```
    $>./print_groups.sh
    god,root,admin,master,nours,bocal$>
    ```

  - for FT_USER=daemon, the result is "daemon,bin" (without quotation marks)

    ```
    $>./print_groups.sh
    daemon,bin$>
    ```

`man id`

`Get inspired by others, do not let them do your job.`

Chapter V

Exercise 02: find_sh

	Exercise 02
	find_sh.sh
Turn-in directory : $ex02/$	
Files to turn in : `find_sh.sh`	
Allowed functions : `None`	

- Write a command line that searches for all file names that end with ".sh" (without quotation marks) in the current directory and all its sub-directories. It should display only the file names without the .sh.
- Example of output :

```
$>./find_sh.sh | cat -e
find_sh$
file1$
file2$
file3$
$>
```

Do not believe any source of information: always make your own tests, controls and verifications.

Chapter VI

Exercise 03: count_files

	Exercise 03
count_files.sh	
Turn-in directory : $ex03/$	
Files to turn in : `count_files.sh`	
Allowed functions : `None`	

- Write a command line that counts and displays the number of regular files and directories in the current directory and all its sub-directories. It should include ".", the starting directory.
- Example of output :

```
$>./count_files.sh | cat -e
42$
$>
```

Failure is part of your learning journey.

Chapter VII

Exercise 04: MAC

	Exercise 04
MAC.sh	
Turn-in directory : *ex*04/	
Files to turn in : `MAC.sh`	
Allowed functions : `None`	

- Write a command line that displays your machine's MAC addresses. Each address must be followed by a line break.

```
man ifconfig
```

```
Collaboration is a key to success.
```

Chapter VIII

Exercise 05: Can you create it ?

	Exercise 05
Can you create it ?	
Turn-in directory : $ex05/$	
Files to turn in : `"\?$*'MaRViN'*$?\"`	
Allowed functions : `None`	

- Create a file containing only "42", and NOTHING else.
- Its name will be :

```
"\?$*'MaRViN'*$?\"
```

- Example :

```
$>ls -lRa *MaRV* | cat -e
-rw---xr-- 1 75355 32015 2 Oct 2 12:21 "\?$*'MaRViN'*$?\"$
$>
```

Milestone Achieved, Keep Going!

You have reached the end of the mandatory exercises to validate this project.

It's up to you to decide if you want to continue with the following optional exercises or switch to your next project. Both paths will make you see useful elements one day or another.

To make your choice please consider the following elements:

- The very first exam is about C programming. So you may have already experienced the very first C project before. Same for the rush at the end of the week (you'll learn soon about the rush).
- Your excellence in this Piscine will be evaluated on multiple factors. The completion of each project is one of them, but the overall progress through the entire list of projects of the Piscine is another. Choose wisely to optimize your results.
- It will always be possible to try the same project again in a couple of days/weeks, until the end of the Piscine.
- Keeping synchronised with your peers ensure a better collaboration.

Chapter IX

Exercise 06: Skip

	Exercise 06
skip.sh	
Turn-in directory : $ex06/$	
Files to turn in : `skip.sh`	
Allowed functions : `None`	

- Write a command line that displays one line out of two for the command `ls -l`, starting from the first line.

`Git push regularly.`

Chapter X

Exercise 07: r_dwssap

	Exercise 07
r_dwssap.sh	
Turn-in directory : *ex*07/	
Files to turn in : `r_dwssap.sh`	
Allowed functions : `None`	

- Write a command line that displays the output of a `cat /etc/passwd` command, removing comments, every other line starting from the second line, reversing each login, sorted in reverse alphabetical order, and keeping only logins between FT_LINE1 and FT_LINE2 included, and they must separated by ", " (without quotation marks), and the output must end with a ".".
- Example: Between lines 7 and 15, the result should be something like this :

```
$> ./r_dwssap.sh
sstq_, sorebrek_brk_, soibten_, sergtsop_, scodved_, rlaxcm_, rgmecived_, revreswodniw_, revressta_
    .$>
```

Rigorously follow the order indicated in the instructions.

Did you check with your left neighbor ?

Chapter XI

Exercise 08: add_chelou

	Exercise 08
	add_chelou.sh
Turn-in directory : *ex*08/	
Files to turn in : `add_chelou.sh`	
Allowed functions : `None`	

- Write a command line that takes numbers from variables `FT_NBR1`, in `'\"?!` base, and `FT_NBR2`, in `mrdoc` base, and displays the sum of both in `gtaio luSnemf` base.
  - Example 1:

```
FT_NBR1=\'?"\"'\
FT_NBR2=rcrdmddd
```

  - The sum is :

```
Salut
```

  - Example 2 :

```
FT_NBR1=\"\"!\"\"!\"\"!\"\"!\"\"!\"\"
FT_NBR2=dcrcmcmooododmrrrmorcmcrmomo
```

  - The sum is :

```
Segmentation fault
```

Do not hesitate to pickup randomly someone in your cluster to ask a question.

Chapter XII

Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double check the names of your files to ensure they are correct.

> ⚠ `You need to return only the files requested by the subject of this project.`